

0.1 Section 4.2 - Binary

Let's compute

$$67^{2025} \pmod{77}$$

Work

Let's see if FLT works

$77 = 7 \cdot 11$ is **NOT** prime, so FLT does not apply

Let's see if EFLT works

1. $77 = 7 \cdot 11$ is the product of 2 distinct primes
2. $\gcd(67, 77) = 1$

So EFLT applies

$$6^{(7-1)(11-1)} \equiv 1 \pmod{77}$$

$$67^{60} \equiv 1 \pmod{77}$$

Breaking up 2025...

$$2025 = 33 \cdot 60 + 45$$

$$67^{2025} \equiv 67^{45} \pmod{77}$$

Break up 45 into powers of 2

$$67 \pmod{77} = 67$$

$$67^2 \pmod{77} = 23$$

$$\begin{aligned} 67^4 \pmod{77} &= 23^2 \pmod{77} \\ &= 67 \end{aligned}$$

$$67^8 \pmod{77} = 23$$

$$67^{16} \pmod{77} = 67$$

$$67^{32} \pmod{77} = 23$$

$$\begin{aligned}
67^{2025} &\equiv 67^{45} \pmod{77} \\
&\equiv 67^{32+8+4+1} \pmod{77} \\
&\equiv 67^{32} \cdot 67^8 \cdot 67^4 \cdot 67^1 \pmod{77} \\
&\equiv 23 \cdot 23 \cdot 67 \cdot 67 \cdot 67 \\
&\equiv 23^2 \cdot 67^2 \pmod{77} \\
&\equiv 1 \pmod{77}
\end{aligned}$$

How do we express 2025 as powers of 2?

Let's list the powers of 2

$$\begin{aligned}
2^0 &= 1 \\
2^1 &= 2 \\
2^2 &= 4 \\
2^3 &= 8 \\
2^4 &= 16 \\
2^5 &= 32 \\
&\vdots \\
2^{10} &= 1024 \\
2^{11} &= 2048
\end{aligned}$$

Now let's break up 2025 using a method similar to the Euclidean Algorithm

$$\begin{aligned}
2025 &= 2^{10} + 1001 \\
1001 &= 2^9 + 489 \\
489 &= 2^8 + 233 \\
233 &= 2^7 + 105 \\
105 &= 2^6 + 41 \\
41 &= 2^5 + 9 \\
9 &= 2^3 + 2^0
\end{aligned}$$

$$2025 = 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^3 + 2^0$$

We can use this to write it in **binary representation**.

$$\begin{array}{cccccccccccc} 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1_2 \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ 2^{10} & 2^9 & 2^8 & 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

Note

The 2 subscript denotes that this is in binary / base 2

Where 1 and 0 represent the coefficient attached to each power. Since it's 1 and 0, it simply represents the power being present and 0 being absent

But notice that if the number n is odd, then the last number of the binary representation is 1

Else if n is even then the last digit will be 0

In other words

$$\begin{array}{l} n \bmod 2 = 1 \text{ if odd} \\ n \bmod 2 = 0 \text{ if even} \end{array}$$

$$\begin{array}{l} 2025 = 2 \cdot 1012 + 1 \\ 1012 = 2 \cdot 506 + 0 \\ 506 = 2 \cdot 253 + 0 \\ 253 = 2 \cdot 126 + 1 \\ 126 = 2 \cdot 63 + 0 \\ 63 = 2 \cdot 31 + 1 \\ 31 = 2 \cdot 15 + 1 \\ 15 = 2 \cdot 7 + 1 \\ 7 = 2 \cdot 3 + 1 \\ 3 = 2 \cdot 1 + 1 \\ 1 = 2 \cdot 0 + 1 \end{array}$$

Writing the sequence of remainders backwards we see

$$\underbrace{1}_{\text{last}} \ 1 \ 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 0 \ \underbrace{1_2}_{\text{first}} = 2025$$

But why does this work?

Given $n \in \mathbb{Z}$

Want $n = a_k 2^k + a_{k-1} 2^{k-1} + \dots + a_2 2^2 + a_1 2 + a_0$

Using the division algorithm,

$$n = 2q_0 + a_0$$

Then

$$q_0 = \frac{n - a_0}{2}$$

Now repeat this with q_0

$$\begin{aligned} q_0 &= 2q_1 + a_1 \\ &\vdots \end{aligned}$$

Let's do an example.

Example 1

Write 1577 in binary

$$\begin{aligned}
1577 &= 2 \cdot 788 + 1 \\
788 &= 2 \cdot 394 + 0 \\
394 &= 2 \cdot 197 + 0 \\
197 &= 2 \cdot 98 + 1 \\
98 &= 2 \cdot 49 + 0 \\
49 &= 2 \cdot 24 + 1 \\
24 &= 2 \cdot 12 + 0 \\
12 &= 2 \cdot 6 + 0 \\
6 &= 2 \cdot 3 + 0 \\
3 &= 2 \cdot 1 + 1 \\
1 &= 2 \cdot 0 + 1
\end{aligned}$$

Writing the bottom to top left to right...

11000101001

We could check this by just doing

$$1577 = 2^{10} + 2^9 + \dots$$

0.1.1 Base n Representation

Y'know, 2 isn't special. We could actually use any number!

Let's try representing 1577 in base 8. Remember that each remainder we get is the coefficient to the corresponding power

$$\begin{aligned}
1577 &= 8 \cdot 197 + 1 \\
197 &= 8 \cdot 24 + 5 \\
24 &= 8 \cdot 3 + 0 \\
3 &= 8 \cdot 0 + 3
\end{aligned}$$

Rewriting this, we get

$$\begin{array}{cccc} 3 & 0 & 5 & 1 \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ 8^3 & 8^3 & 8^2 & 8^0 \end{array}$$

So

$$1577 = 3 \cdot 8^3 + 5 \cdot 8^1 + 1 \cdot 8^0$$

Now let's do it with base 16

$$1577 = 16 \cdot 98 + 9$$

$$98 = 16 \cdot 6 + 2$$

$$6 = 16 \cdot 0 + 6$$

$$\boxed{1577 = 629_{16}}$$

$$1577 = 6 \cdot 16^2 + 2 \cdot 16^1 + 9 \cdot 16^0$$

For our last example, write the $AB0CD_{16}$ in base 10 (aka normal numbers)

Recall that

$$A = 10$$

$$B = 11$$

$$\vdots$$

$AB0CD$ is just 10 11 0 12 13

Since there's 5 "numbers" that means we start with $16^{5-1} = 16^4$

$$\begin{aligned} AB0CD_{16} &= 10 \cdot 16^4 + 11 \cdot 16^3 + 0 \cdot 16^2 + 12 \cdot 16^1 + 13 \cdot 16^0 \\ &= \boxed{700621} \end{aligned} \quad \text{(just plug into calc)}$$